

# Manual

## Batch / Lot Data Management TRT-CLV 8.1

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# 1 Batch / Lot Data Management

## Fields of application

Batch data management provides functions to monitor the production process not only in relation to the recorded quantities and times but also in relation to the used and produced materials.

Validity checks integrated in posting functions increase process reliability by making sure that only admissible batches enter the production process.

As individual events are recorded, it is also possible to research at a later point in time which input materials have been used and which material has been produced.

## Implementation notes

You use the batch and lot management function if you

- are obliged to prove traceability by statutory regulations
- have to ensure traceability as supplier due to (self) commitments
- wish to record and monitor your work-in process stock levels for your multi-level production processes

## Integration

Batch data management is based on order data provided by external systems (mostly ERP systems) or created in the system itself.

Data from external systems (mostly ERP systems) or data created in the system itself may also be accessed to check the material in use for validity.

While planning, discrete batches may be specified for production.

The produced batches can be made available for the number pool of quality data collection.

The data collected with respect to the used materials and output materials can be uploaded to external systems (systems with inventory-management component).

## Features

- Time and quantity recording for batches:
  - Entry of batches at the same time as logging operations on
  - Batch-related time and quantity recording

- Entry and checking of input and output batch changes while the operation is running
  - Manual or automatic assignment of batch numbers according to the configuration
  - Entry of events and logging
  - Entry of notes      entry of notes about batches to be used in label printing and the history
- Checking of input batches
  - Checking of input batches with respect to the duty of documentation when logging operations on as well as to the component list of the operation
- Entry of goods receipt batches
- Configuration options:
  - Editing function: material type master data to control processing specific to material
  - Editing function: to define individual yield, scrap and recording reasons
  - Configuration of process parameters for identification keys such as batch numbers or serial numbers
  - Control of data filing on the basis of configuration parameters → automatic control of data filing on the basis of set configuration parameters
- Editing and correction functions for entered batches
- Evaluation of recorded batch values
  - Tabular presentation of recorded parameters on identification keys
  - Automatic locating and assignment of stored values for evaluations/reports and further processing
  - Transfer of the recorded batch numbers to the ERP system

## 2 Batch data overview

### Overview

|                        |                                                                  |
|------------------------|------------------------------------------------------------------|
| Menu                   | Material management → Inventory management → Batch data overview |
| Transaction code       | batov                                                            |
| Function authorization | batov                                                            |

| Available user fields                                                                               |                            |               |
|-----------------------------------------------------------------------------------------------------|----------------------------|---------------|
| Where                                                                                               | Object type/user field key | Source (type) |
| Detail view                                                                                         | CNR/SYSTEM                 | Batch (MF-D)  |
| <a href="#">How to configure user fields?</a> <a href="#">Which user field types are available?</a> |                            |               |

### Purpose

Use the *batch data overview* to view or edit one or more batches depending on the entered selection criteria.

### Integration

The *batch data overview* shows all existing batches matching the entered selection criteria.

The *batch data overview* shows the selected batches and their current status including further details.

### Selection criteria

The application provides the following selection criteria:

#### "Batch" category

##### Batch number

The batch number represents the batch ID used by the user. The user can:

- enter/view the batch number in terminal dialogs
- use the batch number for tracing or information search via the office client



If the configuration option "automatic generation of the batch number" is enabled, leave the "batch number" field empty in the editing dialog when you add a new batch. The system will only assign the batch number automatically, if the "batch number" field is empty.

**Material**

Enter the material number as a selection criterion to display all batches assigned to this material number.

**Workplace**

Enter the workplace as a selection criterion to display all batches produced at this workplace/machine.

**MES order number**

MES order number (order/operation number) that produced the batch.

**Internal batch number**

The batch number is a **unique, system-wide** batch identification number.

**Material buffers**

The individual batches are located in a specific [Material buffers](#). Enter the material buffer to view all batches assigned to the selected material buffer.

**Material type**

The individual batches or materials belong to a [Material type](#). The same transport and handling guidelines apply to these material types across the system. Enter the material type to view all batches assigned to the selected material type.

**Material category (kind of material)**

The material category assigns batches to specific classes/ groups. Enter the material category to view all batches assigned to the selected material category.

**Material name**

The system takes over the material name from the currently logged in order when output batches are generated at shop floor terminals. As no material master is managed, the material name is stored redundantly in the batch description.

**Manufacturing date from / until**

Date/period of the production of a batch.

**Consider long-term data**

You can also select archived data.

**Historic batch entries**

If you use the function *historic batch entries* and you also use throughput batch numbers or serial numbers, the application will also show batches with the status "processed" for every throughput batch number or serial number. Use the function *historic batch entries* to get further process information on a throughput batch number or serial number.

**Status category****Batch class**

The batch class describes the overall quality of the batch. Enter the batch class to view all batches matching the selected batch class.

**Batch status**

The batch status describes the technical system and production status of a batch. Enter the *batch status* as a selection criterion to view all batches assigned to this status.

**Quality status**

The quality status "blocked" prevents a batch from being logged on. Enter the quality status as a selection criterion to view all batches assigned to this quality status.

**Manual Q status**

Enter the manual Q status as a selection criterion to view all batches assigned to this status.

**Material status**

The material status indicates a logical status of the batch, e.g. packed, tested. Enter the *material status* as a selection criterion to view all batches assigned to this material status.

**Transport status**

The transport status represents the technical system status with respect to transfer postings to external storage. Select the transport status as a selection criterion to display all batches that are assigned to this transport status.

**"Advance logon" option**

Use this option to restrict the data to [batches logged on in advance](#).

**"Attributes" category****Attribute (1 to 10)**

You can use the attributes directly stored with the batch to restrict the data displayed.

**"Batch attributes" category****Batch attribute (name)**

You can use the [batch attributes](#) configured for the material type to restrict the data displayed. There are 40 text fields, 20 numeric fields and 20 decimal fields that may be configured.

**"Alternative batch numbers" category****Alternative batch number (1 to 20)**

Select an alternative batch number as a selection criterion to display all batches that are assigned to this number.

**"Reservation" category****Reserved for order**

Enter or select the order number in this field to display all batches that were produced for this order/ OP.

**Reserved for OP**

Enter or select the order/OP number in this field to display all batches that were produced for this order/ OP.

**Dates category****Expiry date from / until**

Date/period that indicates the shelf life of a batch.

**Availability date from / until**

Date/period of the availability of a batch.

**Warning time**

Warning date of a batch

**"Miscellaneous" category****Serial number**

Enter a *serial number* as a selection criterion to display all batches that are assigned to this serial number via the *serial number* field.

**Batch number**

Select the batch/lot number as a selection criterion to display all batches that are currently assigned to this number.

**Person**

Select *person* as a selection criterion to display all batches that have been produced by this person and, as a result, are currently assigned to this person (personnel number).

**Collective batch (merged batch)**

Enter a *collective batch number* (merged batch number) as a selection criterion to display all batches that are assigned to this collective batch number in the *collective batch* field (*merged batch* field).

**PPS batch**

Enter a *PPS batch* as a selection criterion to display all batches that are assigned to this PPS batch number in the *PPS batch* field.

## Editing functions

Use the following functions that are available in addition to the standard features to edit one or several data records:



### Add batch

Use this function to insert a new batch. Goods movements are not generated for the new batch.



### Copy batch

Use this function to copy a batch. However, you have to enter a new batch number for the new batch. Goods movements are not generated for the new batch.



### Edit batch

Use this function to edit a batch. But neither cancellations nor goods movements are generated for the changed batch.



### Delete batch

Use this function to delete a batch. But neither cancellations nor goods movements are generated for the deleted batch.



### Merge batches (new batch number)

Merging batches. A new batch number is generated for the new batch ([Merge batches](#)).

When merging collective batches, all individual batches are assigned to a new collective batch number.



### Merge batches (use existing batch numbers)

Merging batches. A batch number already included in the merged batches is used for the new batch ([Merge batches](#)).

When merging collective batches, all individual batches are assigned to the selected collective batch number that is part of the merger.



### Split batch

Create a new batch by splitting it off from an existing batch ([Batch split](#)). In this case, you can choose from the following options to deal with the remaining target quantity of the original batch:

- Repost the remaining quantity of the batch to the new batch
- Reduce the existing batch to the remaining quantity

If you split a collective batch, the selected individual batches/serial numbers are transferred to a batch split off. The remaining individual batches/serial numbers remain assigned to the original collective batch.



### Repost

Use this function to repost a batch to another material buffer ([Repost batch](#)).



### Generate

Use the function "enter goods receipt batch" to create batches manually ([Generate batch](#)). This is necessary if material is delivered via the incoming goods, for example.



### Edit batch attribute

The tab "batch attributes" shows the [batch attributes](#) of the batch selected in the grid. The application only shows those batch attributes that have been assigned the option "show attribute on client" within the configuration of [batch attributes](#).

Provided that [batch attributes](#) are shown, click the button "edit batch attribute" to edit the batch attributes. In this case, the system does not check the field types/field lengths specified in the configuration of [batch attributes](#).

## Go to



### Graphic batch tracing

Calls up the application [Graphic batch tracing](#).



### Document management

Calls up the [document management](#)

## "Batch data overview" detail application

The *batch data overview* detail application shows all batches matching the entered selection criteria.

The documents entitled [batch object](#) and [batch structure](#) outline the information or field descriptions of the selected batches and, thus, of the detail application.

## 3 Material Types

### Overview

|                        |                                                    |  |
|------------------------|----------------------------------------------------|--|
|                        | →                                                  |  |
|                        | →                                                  |  |
| HYDRA menu             | Master data → Material → Material types            |  |
| FEDRA menu             | Detailed scheduling → Master data → Material types |  |
| Transaction code       | mtyp                                               |  |
| Function authorization | mtyp                                               |  |

| Available user fields                                                                               |                            |                      |
|-----------------------------------------------------------------------------------------------------|----------------------------|----------------------|
| Where                                                                                               | Object type/user field key | Source (type)        |
| Detail view                                                                                         | MATTYP/SYSTEM              | Material type (MF-D) |
| <a href="#">How to configure user fields?</a> <a href="#">Which user field types are available?</a> |                            |                      |

### Purpose

Use this function to create or modify material types in the system.

### Integration

The system uses the material type and the settings stored with it to control the system behavior of objects that are assigned to this material type.

### Selection criteria

#### Material type

Unique material type key

#### Type

Category of the material type

#### Description

Description or name of the material type (plain text)

### Field descriptions - General

#### Material type

Unique material type key

#### Batch size

Typical batch size for batches of this material type.

**Description**

Description/name (plain text) of the material type

**Type**

Category of the material type

You can assign a type to group and classify material types that belong to the same category. Assigning a type, however, does not control the processing.

**Unit**

Unit used for materials of this type.

**Field descriptions – Input batch processing****Inventory management**

**E** - Yes, when logging input batch off

Here, you manually record the used quantity of a batch or the remaining quantity when you log off the batch. Once the batch is logged off, quantities are reduced and the consumption is uploaded to the higher-level systems.

**R** - Yes, backflush (retrograde)

Once the batch is logged off and the operation is interrupted or logged off, the consumption values are reported to the higher-level system.

**G** - Yes, backflush (only with output batch = yield)

Same procedure as with R; but the quantity of the input material is only reduced in a retrograde manner if the output batch produced is a yield quantity (batch class = G).

**N** - No inventory management. Batches of this material type are not included in the stock overview.

Note: Depending on the configuration, you can enter the remaining quantity or the consumption. In general, you can do so, when changing input batches via the Windows terminal. When logging off the input batch, this quantity is posted to the respective batch. This applies for options E, R, G and N.

**A** – Anonymous inventory management

Batches of this material type are processed as material without batch reference when it comes to composition.

**Log off input batches (when OP is interrupted/logged off)**

**A** – Input batches are automatically logged off when an operation is interrupted or logged off and then get the batch status "processed".

**F** – Input batches are automatically logged off when an operation is interrupted or logged off and then get the batch status "free".

**G** – Input batches are logged off when an operation is interrupted or logged off according to the configurations in the basic parameter settings.

**N** – Input batches are not automatically logged off when an operation is interrupted or logged off.

**S** – Input batches are automatically logged off when an operation is interrupted or logged off and then get the batch status "blocked".

### Tolerance when logging off

Value in %.

When the input batch is logged off, the system checks if the absolute value of the remaining quantity is less than the percentage value of the original quantity specified in this field. If this is the case, the batch is set to batch status *Processed* and identified as consumed.

### Maximum value

The maximum value specifies the tolerance when logging off.

If the remaining quantity is less than the specified percentage value but greater than the maximum value, the batch is not set to batch status *Processed*.

### Standard quantity of unknown input batches

If a quantity for unknown input batches is specified and an unknown input batch of this material type is recorded, this unknown input batch will be entered with the specified quantity. If no value is specified, the system generates the unknown input batch with the quantity 1,000,000,000.

### Options

#### Can be logged on several times to one machine

You can use this option to log on an input batch to a machine for multiple items of the component list (BOM items) of an operation.

If the option is not set, the system issues an error message if you try to log on an input batch that is already logged on once more.

Example:

The following material components are assigned to an operation:

|                      |                        |
|----------------------|------------------------|
| BOM item <b>0001</b> | Material <b>487191</b> |
| BOM item 0002        | Material 923948        |
| BOM item <b>0003</b> | Material <b>487191</b> |
| BOM item 0004        | Material 574857        |
| BOM item <b>0005</b> | Material <b>487191</b> |

If the option is enabled, batch 000007153 with material number 487191 can be logged on to one of the three items 0001, 0003, 0005 or to two or even to all of these items. If the batch is already logged on, the system shows a selection window every time you attempt to log on the batch once more. The window shows the available BOM items of the material. To log on the batch once more,

the user has to select the required item. After confirmation, the batch is logged on to this (additional) item of the BOM. If the user selects a BOM item where the batch is already logged on, the labeling of the button changes and the respective batch can be logged off.



In general, different batches can be logged on several times for the same material (same material and different BOM items), irrespective of how this option is set. If this option is enabled, the system assumes that the option *Can be logged on to several machines at the same time* is also enabled.

This option does not allow to log on unplanned material.



In case of merged batches, you cannot use the option *Can be logged on several times to one machine*.

### Can be logged on to several machines at the same time

You can log on the input batch to several workplaces and/or operations at the same time. In this case, it does not make any difference if the input batch that is already logged on to an operation at a machine

- is logged on to another operation of the same machine,
- is logged on to another operation of another machine,
- is logged on to the same operation of another machine.



In case of merged batches, you cannot use the option *Can be logged to several machines at the same time*.

### Input batch must be logged on

This option activates an additional validation check when an operation is logged on that requires batch management. For all material components (BOM items) with a material type where this check is enabled, the system checks if batches have already been logged on. If the check has a negative result, the logon of the operation is refused.

If you want all materials of the component list to be logged on as input batches before the OP can be logged on, you can configure this via the material type. To do so, use the option *Log OP on only if all components are logged on as inp. batches* of the tab *Output batch processing*.



The system performs the validation check **ONLY** when you log on the operation.

When an input batch is changed using the function *Input batch change*, no check takes

place.

That means: In case of an input batch that must be logged on (see this option), you can log off this input batch, without logging on a new batch for the respective BOM item.

### Hand batch number down

If you log on an input batch with a material type where this option is set when the order starts, the system uses this batch number as batch number for the output batch to be logged on.

At the same time, the system checks if the batch number of the input batch may be handed down when the output batch is logged on, i.e. the system checks if the option is enabled for the material type of the input batch.

This option is processed and [run-through batch processing](#) in combination with the [configured](#).



If this option is set, you also have to set the option "Input batch only valid for exactly 1 output batch".

### Input batch only valid for exactly 1 output batch

If this option is enabled for the material type of a component, you must log on at least 1 batch as input batch for each output batch.

Note: This option only controls how the batch change is processed (CA\_WL). This option does not ensure that an input batch can be logged on again at a later point in time (when another output batch is active).

This option is processed and [run-through batch processing](#) in combination with the [configured](#).



If you use this option without run-through batch processing, incorrect batch assignments can occur during tracing. In this case, the succeeding input batch is linked to the output batch that must actually be logged off.

### Allow entry of unknown input batches

Using this option, the system can automatically generate and log on input batches that the system didn't know beforehand.

If you enable this option and if the input batch is not known in the system when you log on the input batch, then the system creates this unknown batch using the selected BOM item (material). The batch is created with a quantity of 1000000000 in primary quantity unit of the operation and a goods receipt posting is performed.

To identify the correct material type, a configuration/assignment between [material and material type](#) must be available so that the system can uniquely identify the material type using the material number of the component. If a material is not assigned to a material type, the material type *SYSTEM* is automatically assigned.



Requirement: You must activate the recording of unknown batches in the basic settings using the option *MPL → Create unknown batches*.

### Use data for output material

This option specifies if the system uses the expiry date of the input batch to calculate the expiry date of the output batch. Note: This option is only available if a logged on input batch has a material type where this option is enabled. Only in this case the generated output batch can use the expiry date of the input batch if configured accordingly (see "Determination of expiry date" in the "General" tab).

### Decision on changing the input batch

If this option is enabled, the system displays a warning message when the input batch is changed and before the current input batch is logged off.

Especially when serial numbers are collected, this warning message informs the user that all components with a serial number must be evaluated before logging the input batch off. Only then, the components can be "properly" linked to the resulting output batch (merged batch).

### Pass batch attributes on

You can use this option to transfer configured batch attributes of an input batch to the resulting output batches. The configuration can be complemented flexibly using the extended object configuration. The configuration is described in the following document: [Configuration to pass on batch attributes](#)

### Pass document links on

You can use this option to transfer document links of an input batch to the resulting output batches. The configuration can be complemented flexibly using the extended object configuration. The configuration is described in the following document: [Configuration to pass on document links](#)

## Field descriptions – Output batch processing

### Min. storage time

Minimum storage time of the material type

The system uses this value to calculate the availability date when the output batch is generated (by adding it to the manufacturing date/time). As long as this calculated point in time is in the future, a batch remains in "Min. storage time" status and cannot be logged on.

Note: You can use the configuration [Assignment material to material type](#) to override this setting for specific material.

### Warning limit

Warning limit of the material type

The system uses this value to calculate the respective warning date when the output batch is generated (by adding it to the manufacturing date/time). You can use the function [Warning report](#), for example, to integrate this value in evaluations.

Note: You can use the configuration [Assignment material to material type](#) to override this setting for specific material.

### Expiry limit

Expiry limit of the material type

The system uses this value to calculate the respective expiry date when the output batch is generated (by adding it to the manufacturing date/time). At this point in time, the batch is automatically set to "Expired" status and can no longer be logged on.

Note: You can use the configuration [Assignment material to material type](#) to override this setting for specific material.

### Reserved when batch is generated

You can use this option to automatically reserve the output batches produced for subsequent processes. You can make reservations on different levels.

Values:

- - No automatic reservation when batches are generated
- - Reservation for all OPs of the order
- - Reservation for subsequent OPs

With the last two values, the system checks if a reservation is available when an input batch is logged on subsequently.

In coil-based production (MPL-RF) and with so-called cutting machines, the reservation is always made for the subsequent mother operation.

### Options

#### Input batch change mandatory if remaining qty. $\leq 0$

If this option is enabled, the currently logged on input batches are highlighted in red that have a remaining quantity, which is less than or equal to 0, when an output batch is changed. These input batches must now be logged off.



This option is only applicable for MPL machines (batch management) with machine type "M". The terminal processes this option locally.



This option is only applicable for the dialog *Output batch change* (CA\_WL\_MPL). The remaining quantity displayed for the logged on input batch must be equal to or less than zero ( $\leq 0$ ). The system can only perform this check, if the remaining quantity of the input batch is less than or

equal to 0 ( $\leq 0$ ) before the posting.

**Log OP on only if all components are logged on as inp. batches**

If this option is set, the OP using the output material can only be logged on if all materials of the component list are logged on as input batches.

If this option is not set, the OP can be logged on even if not all of the materials are logged on as input batches.

**Consumption balance**

When you log off an OP, the system opens an additional dialog (V\_BLZ) displaying the material components and their consumption quantities in relation to the OP that is currently logged on. In this dialog, you can also log off input batches that are still running. The option only becomes active when the consumption balance of the machine has been activated in the "MPL" tab.

**Serial no. requirement (MPL-SNR, as of MPL 7.2)**

If the system records serial numbers (ADE-SNR), and if this option is active and the operations require batch management, then the system generates an additional batch per serial number in the MPL module. Batches are only collected per serial number (using command A\_TR) if the operations require serial number management = E. In the table "Batch relationships" (mpl\_beziehungen), traceability to the current output batch (ID) is established using the serial number recorded.

**Generation of HU (MPL-SNR, as of MPL 7.2)**

If the option "Serial number requirement" is set, you can use this option to control if the serial numbers recorded are stored in relation to a handling unit (merged batch). The option is currently only intended for customizations and does not have any effect.

Note: The system only checks this option when the OP is logged on and when the output batch is changed. When an input batch is changed using the function *Input batch change*, no check is performed.

**Log off/interrupt OP without the last batch**

If the option is set, a last output batch is generated for the article of the operation (matching this material type) between the last output batch change and/or OP logon and the OP interruption and/or OP logoff, but this output batch is immediately deleted (batch status = "D"). The batch is not visible in the batch history. For further information on this function, refer to the document ["Interrupting/Logging off OP without the Last Batch"](#).

**Delete batch assignment**

If this option is enabled, the link between the output batch "deleted last" to the currently running input batches of the OP is deleted. As a result, the deleted batch is not visible in batch tracing.

### Automatic assignment of serial numbers

This option enables the automatic assignment of numbers for a new part when merging components listed by serial numbers. If this option is not set, you can assign the serial number manually. The option is only relevant if the option "Superordinate serial number" = N is set for the component.

### User field key for output batches

The defined user field key is transferred to the generated output batches.

### Behavior when output batch has quantity 0

Use the option *Behavior when output batch has quantity 0* to specify system behavior when output batches are changed and the quantity is 0. You can choose from the following three configuration options.

- |                        |                                                                                                                                                              |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>"empty"</b>         | No reaction. Output batches are changed with quantity 0.                                                                                                     |
| <b>"Warning"</b>       | A warning message is displayed. You can still override the warning message. If you override the warning message, output batches are changed with quantity 0. |
| <b>"Error message"</b> | An error message is displayed. The error message cannot be skipped. The output batches cannot be changed with quantity 0.                                    |

### Standard quantity of output batches

If a quantity is entered for output batches and an output batch of this material type is generated, the *quantity* field of the shop floor client is populated with the specified quantity. You can still change this default assignment at any time.

## Field descriptions - General

### Determination of expiry date

- **Current point in time**  
The expiry date is calculated using the current point in time and the duration stored for the material type of the output batch.
- **Shift date**  
The expiry date of the generated output batch is identified using the current shift date, the time 00:00 and the expiry limit stored for the material type.
- **Input batch**  
The expiry date of the generated output batch is identified using the running input batch where the option "Use data for output material" is set for the material type.
- **Processing based on enhanced object configuration**  
An additional configuration in the [Advanced object configuration](#) specifies how the expiry date of the generated output batch is identified:

○ **Use the closest expiry date of all logged on input batches**

If this configuration is active, the expiry and warning time of the input batch with the closest expiry time is used for the output batch. The availability date is calculated using the configuration in the "Output batch processing" tab.

| Parameter name  | Parameter value                                                                       |
|-----------------|---------------------------------------------------------------------------------------|
| Object type     | MATTYP                                                                                |
| Object ID 1     | Material type of the output batch. The configuration is valid for this material type. |
| Parameter       | DATE_OF_EXPIRY_CALCULATION                                                            |
| Parameter value | MIN_DOE_OF_INPUT_MATERIAL                                                             |

○ **Use the closest expiry date of selected input batches**

If this configuration is active, the expiry and warning time of the input batch with the closest expiry time is used for the output batch. To identify the closest expiry date, the system only uses input batches of the configured material type. The availability date is calculated using the configuration in the "Output batch processing" tab.

| Parameter name                                                                                                                                                 | Parameter value                                                                       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Object type                                                                                                                                                    | MATTYP                                                                                |
| Object ID 1                                                                                                                                                    | Material type of the output batch. The configuration is valid for this material type. |
| Parameter                                                                                                                                                      | DATE_OF_EXPIRY_CALCULATION                                                            |
| Parameter value                                                                                                                                                | MIN_DOE_OF_SELECTED_INPUT_MATERIAL                                                    |
| When the relevant input batches are identified, the system only uses the batches that can be identified via an additional configuration for the material type. |                                                                                       |
| Parameter name                                                                                                                                                 | Parameter value                                                                       |
| Object type                                                                                                                                                    | MATTYP                                                                                |
| Object ID 1                                                                                                                                                    | Material type for which the configuration applies                                     |
| Parameter                                                                                                                                                      | DATE_OF_EXPIRY_CALCULATION                                                            |
| Parameter value                                                                                                                                                | INCLUDE_IN_DOE_SELECTION                                                              |

### WIP material (Work in Process)

This option specifies whether a batch of this material type is automatically logged on as input batch for the subsequent operation (according to the operation sequence of a production order) (as [WIP material](#)). Find a description of how to configure this function [here](#).

## Options

### Transfer to interface (goods movements)

If this option is enabled, the produced output batch is identified as goods receipt or the used up input batch as goods issue. These [material movements](#) must then be uploaded.

### Hold back goods receipt until usage decision is made

You can use this option to delay the transfer of goods receipts to the PPS system. In this case, the output batch is transferred only if a usage decision is taken.

## Field descriptions - Transport

### Generate transport order for output material

If this option is enabled, the system creates a [transport order](#) for a batch when an output batch is produced. The transport starts from the material buffer that includes the output batch. This option overrides the configuration of the respective option in the workplace configuration.

### Generate transport order for input material

Use this option to create a [transport order](#) for an article (and relating to a material component) when you plan an operation at a machine using the shop floor scheduling module. Transport starts from the output material buffer of the preceding operation. This option overrides the configuration of the respective option in the workplace configuration.

## Field descriptions – User fields

You can customize the system to release user fields for the object type "MATTP".

## 4 Reasons

### Summary

|                        |                                               |
|------------------------|-----------------------------------------------|
| Menu                   | Master data → Workplaces / Machines → Reasons |
| Transaction code       | reas                                          |
| Function authorization | mdreas                                        |

### Usage

Use this configuration to create or to change the reasons available in the system. Reasons may either be created for the entire system or referred to a workplace

### Integration

The reasons that are saved to the system will be available for collection on the terminal as well as in different applications. They are used to classify quantities of materials or modifications.



In order for the settings or the changes made to be able to be interpreted by the terminal shop floor program, the terminal, which the workplace/machine is assigned to, has to be restarted. All terminals should be restarted, provided that new reasons have been created or reasons affecting the entire system have been changed.

### Requirements

You have defined [reason texts](#) in the system.

### Selection criteria

The following selection criteria are available in the application:

#### Type

Reason type, e.g. scrap

#### Workplace

Workplace selection.



Reasons that are configured for the "workplace" SYSTEM, will always be displayed even if a workplace will explicitly be restricted.

#### Reason

Unique reason number

## Designation

Designation of the reason. Wildcards can be used.

## Superior reason

Selection of a superior reason. All reasons will be selected that have the selected reason as (direct) superior reason.

## Field descriptions

### Workplace

Assignment of a reason text to a workplace. If "SYSTEM" is entered, this will apply as system-wide assignment.



System-wide reasons will always apply in addition to the workplace-specific reasons and will therefore also be displayed in the terminal's selection list.

### Type

Classification and/or grouping of reasons

Possible values:

|   |                                                                  |
|---|------------------------------------------------------------------|
| A | Scrap reason                                                     |
| N | Rework reason                                                    |
| P | Open quantity reason (before: problematic quantity)              |
| G | Yield reason: will be interpreted as deviation reason            |
| L | Reasons for batch logs (relevant in connection with MPL)         |
| R | Reduce (partitioning) reason (relevant in connection with WRM)   |
| E | Increase (partitioning) reason (relevant in connection with WRM) |

### Reason

Identification number of the reason.



As system-wide reasons always apply in addition to reasons relating to workplaces, their numbers have to be unique, i.e. a scrap reason with the number 99 for the SYSTEM workplace must not be defined at the same time as workplace-related scrap reason assigned to the number 99.

### Reason text no.

Identification number of the [reason text](#)

### Designation

Related reason text from the reason text configuration.

**Ext. reference**

For each assignment exists an alphanumeric representation that can be uploaded back to the interface, for example

**Scrap material**

Is used in connection with HYDRA-MPL

**Superior reason**

The reference to a superior reason is reserved for further extensions/modifications; at present it has no function and should consequently not be completed.

**“Copy“ detail application**

The "copy" button can be used to copy reasons defined in relation to a workplace from one workstation to the next. Reasons of the "workplace" SYSTEM cannot be copied.

The below-mentioned options are supported while copying:

- Copy currently selected reason

This option can be used to copy the currently selected reason. For this purpose, enter the below pieces of information in the fields below "To":

- Workplace: target workplace for which the reason is to be copied
- Type: Choose the reason type under which the reason is to be created for the target workplace. The field is assigned by default to the type of the currently selected reason.
- Reason: Enter the reason number under which the reason is to be created for the target workplace. The field is assigned by default to the type of the currently selected reason.

- Copy all reasons

This option allows copying of all reasons defined for a workplace to another workplace. However, a prerequisite for this is that reasons have not yet been configured for the target workplace. To do so, enter the target workplace for which the reasons are to be copied in the "workplace" field. Please note that all workplace reasons are always copied, irrespective of the type of the reason.

- Copy missing reasons

In contrast to the previous option, this function allows for reasons to be copied to another workplaces, even if reasons are already assigned to this workplace. To do so, enter the target workplace for which the reasons are to be copied in the "workplace" field. Please note that all workplace reasons will always be copied, irrespective of the type of the reason.

# 1 Batch Logs

## Summary

|                        |                                                  |
|------------------------|--------------------------------------------------|
| Menu                   | Material management → Batch tracing → Batch logs |
| Transaction code       | batlog                                           |
| Function authorization | batlog                                           |

## Usage

This function is used to display the recorded log data with respect to a batch and evaluate it accordingly. No provisions are made for modifying existing log data.

## Selection criteria

The following selection criteria are available in the application:

### Batch

Batch number

### Order/ OP

Order number or combined order/ OP number in order for all logs within an order to be displayed.

### Machine

Machine/ workplace in order for all logs for a specified machine to be displayed.

### Reason

Number of the reason for the error according to configuration.

### Recording time

Period in which the log data are to be selected.

## Field descriptions

### Batch

Batch for which the log record was recorded.

### Internal batch number

Internal batch number for which the log record was recorded.

### Order

Order during which the log record was recorded.

### Machine

Machine at which the log record was recorded.

**Reason, reason texts**

Recorded reason as well as reason text according to configuration

**Comment**

Recorded comment.

**Running meter from/to**

Running meter position/ area for which the log record was recorded.

**Personnel number, date**

Personnel number of the person that recorded the log record as well as the date on which the log record was recorded by the worker.

**Originator, date**

Most recent originator of the log record as well as the date of the most recent modification.

**Scrap**

Reserved.

**Error message was generated**

Reserved.

**Reference**

Reserved.

**Daughter reels (optional)**

Reserved.

**Attribute 1-5, parameter 1,2**

Reserved for customer-specific additions

## 5 Material Movements

### Summary

|                        |                                                                 |
|------------------------|-----------------------------------------------------------------|
| Menu                   | Material management -> Inventory management ->Material movement |
| Transaction code       | mmov                                                            |
| Function authorization | mmov                                                            |

### Usage

In the evaluation/report of material movements, the various movements related to a material inventory are represented. The material movements can be related to batches or discrete material components.

For movements that are batch-related, additional data from the referenced batch inventory (e.g. alternative batch numbers) are shown in the evaluation.

Material movements are differentiated into goods issued (consumption) and goods received (material produced from production).

### Selection Criteria



If several selection criteria are used overlapping results are displayed.

The application provides the following selection criteria:

#### Movement

##### Movement

Type of movement

##### Movement

Type of movement

##### Batch number

With this selection, all movements of batches assigned to the selected (external) batch number are displayed.

##### Material

Material number of the batches.

**Workplace**

Selection of a workplace from the workplace pool

**MES order number**

MES order number (order/operation number) by which the batch has been produced/consumed.

**Internal batch number**

With this selection, all movements of batches assigned to the selected internal (system-wide unique) batch number are displayed.

**Material type**

With this selection, all movements of batches assigned to the selected material type are displayed.

**Date**

Period during which the movements took place

**Movement**

Type of movement (goods receipt or goods issue)

**Consider long-term data**

Archived data is taken into account.

**Storage locations****Source buffer (storage location)**

Material buffer from which the batch/material was moved/transferred.

**Issuing storage location**

Storage location (from material type configuration) from which the batch/material was moved/transferred.

**Target buffer**

Material buffer which the batch/material was moved/transferred to.

**Receiving storage location**

Storage location (from material type configuration) which the batch/material was moved/transferred to.

**Attributes****Attributes 1 - 10**

Selecting an attribute as a selection criterion displays all movements of batches that currently have this identifier.

**Batch attributes**

**Batch attributes**

Selecting a batch attribute as a selection criterion displays all movements of batches for the material type of which this batch attribute has been configured and that are currently assigned this identifier.

**Alternative batch numbers****Alternative batch number 1-20**

Selecting an alternative batch number as a selection criterion displays all movements of batches that currently have this identifier.

**Field Descriptions****Movement**

Type of material movement

**Uploaded**

If the identifier is set, the material movement has already been uploaded to the ERP system. If the identifier is not set, the material movement has not yet been uploaded to the ERP system.

**To be uploaded**

If the identifier is set, uploading the material movement to the ERP system is intended. This setting can be accessed using the corresponding configuration on the [material type](#). If the identifier is not set, uploading the material movement to the ERP system has not been considered.

**Article**

Material number

**Description**

Material description (plain text)

**OP**

Operation of the material movement

**Issuing storage location**

Original storage location

**Receiving storage location**

Receiving storage location of the material movement



For goods issues quantities are always indicated with negative algebraic sign.

